

NGDR 1500

TECHNICAL SPECIFICATION



- ☐ Fuel efficient
- ☐ Low maintenance

- ☐ Easy to operate
- ☐ Low operating cost

600 Meter Drilling depth capacity (capable to work in sand, clay, black soil , Hard Rock, Boulder or any Types of formation).

Note: The images provided here are for reference only and do not depict the exact specification outlined in this document.



Application

Water Well Drilling, Direct Rotary

Capacity

600 meter @ 24" Dia.
 350 meter @ 28" Dia.
 200 meter @ 42" Dia.
 600 meter @ 6.5" Dia. (DTH)
 400 meter @ 8" Dia. (DTH) (capable to work in all types of formation).

Mast (With hydraulic lifting & lowering system)

12 meter length, 100mm X 100mm ID, structural tubular steel. (square).
 8 mm wall thickness.
 With heavy duty Support pipes and cross supports.
 Lifting Pulley 11 nos. (Cross-over pulley placement for saving time in changing the drill rods and ease of installing the casings).
 Gross load capacity: 100,000 kg.
 Hook load capacity on main travelling 8 sheaves+ fast line: 70,000 kg.
 Hook load capacity :secondary travelling 4 sheaves+fast line: 28,000 kg.

Machine Platform

Dimensions : 8meter length, 2.2 meter width
 Main chassis : SKID - 300mm main channel welded with heavy support structural beams with support structure 150x70 tubular pipes.
 Openable at the site to 3.3 meter width with suitable manually expandable back platform
 Flooring : 6 mm MS Diamond / Checkered Plate
 Design Features : Stable, Vibration Resistant, Suitable for Welding Casing Pipes Directly on the platform.

Water Pump

Triplex Plunger Pump - 200 LPM Dth water injection
 Foam pump for Sticky clay formations
 4x4 Centrifugal Pump for mud mixing

MUD PUMP

6 x 10 RECIPROCATING Dual acting duplex mud pump fitted with separate 6 speed gear box and 6 sheave V-belt pulley drive.

Rotary Table

Hexagonal floating spindle for 7 ¼ ".
Hexagonal key spindle for rotating the Kelley,
5 ½" of Kelley spindle made by forged EN9 material heat treated,
Easy installation of 5" API Drill Pipes connections OD 165mm.
Retractable and removable for lowering of the casing pipes.
Hydraulic movement facility, dual jack.

Prime Mover

ASHOKLAYLAND 320 HP with ZF 9 Speed Pneumatic Gear Box.

Power tuck with 1 input 7 output with eccentric adjustable heavy-duty triplex chain 3 of 7 outputs are joint with gear boxes for the optimum performance and speed.
These gearbox are been made with the heavy-duty 1.25" triplex chain and sprockets. Eccentrics are for the adjustment of the chains for long duration of the chains and sprockets.

Winch(Draw works)

2 NOS of 28" Diameter of draw works driven by crown pinion heavy-duty shaft drive. Set of 16 Nos of friction facings having 20" of clutch diameter, 6 INCH BELT BREAKS ON EACH DRUMS
(AIR VANTILATED BREAKS)

Steel Wire Rope

For main Drum: 20 mm thickness steel core. (6X36)
For Rod line Pulling drum: 20 mm thickness steel core. (6X36)

Drilling Platform

N.G standard made with MS checkered plate of 6 mm thickness,
Plate form for operator with lights and driller seat.
3 sided expandable for easy handling of the drilling operations.
Back platform easy to assemble for welding the casing pipes directly on the machine platform.

Kelly

Type : Hexagonal
 Size : 5 ½" (API)
 Length : 9 meter

Traveling

For Drilling Lifting: 4 nos. Pulley
 : 8 sheave + 1 fast line sheave.
 For Rod Line Pulling : 2 nos. pulley
 : 4 sheave + 1 fast line sheave.

Generator

400 AMP. Suitable for 4mm of welding electrodes.
 Single phase power output for Grinding and lighter applications.

Water Swivel

NG make heavy duty joint with travelling and Kelley.
 Hook Load Capacity : 70,000Kgs, 500 PSI

Paint job

Industrial Grade, Color choice on customer.

Lights

On the plate form, On the mast, At the drilling hole, under the platform.

**Combinations will vary according to our Design feasibility's*

ACCESSORIES PROVIDED WITH MACHINE FREE OF COST		
1	SPANNERS FOR MUD PUMP	3 NOS
2	HOISTING PLUG FOR THE DRILL RODS	1 NOS
3	DRILLER SEAT	1 NOS
4	DRAG BITS 12", 14", 16", 18" EACH	1 SET
5	KELLEY PIN (THREAD PROTRACTOR)	1 NOS
6	RUBBER GLANDS FOE WATER SWIVEL	100 NOS
7	RUBBER GLANDS FOR MUD PUMP	1 SET
8	MUD HOSE R2	1 SET
9	MUD HOSE PU 4"	1 SET
10	MUD HOSE PU 3"	1 SET
11	FOOT VALVE	1 NOS
12	BREAK OUT CYLINDER	1 SET

